Exploratory Computational Research Report

Planetary-Embedded Multi-Agent Systems (PEMAS)

A Synthetic Experimental Framework for
Ecologically Adaptive AI Governance

Environmentally Situated Cognition,
Adaptive Governance,
and Bounded Computational Metabolism

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github.com/MinervaRose/pemas-framework

Abstract

Modern AI systems are often discussed as if computation were abstract and materially detached. In practice, AI systems depend on electrical grids, cooling systems, hardware infrastructure, water, minerals, and finite planetary resources. This report introduces Planetary-Embedded Multi-Agent Systems (PEMAS), a synthetic experimental framework in which ecological and planetary-state variables function as governance signals inside a modular multi-agent AI architecture.

The framework compares a conventional baseline multi-agent pipeline with a governed PEMAS pipeline. Both systems use the same synthetic task set and simulated agent components. The critical architectural difference is that PEMAS allows planetary-state variables to regulate recursion depth, token budgets, routing behavior, compression frequency, and stopping decisions.

The framework is intentionally synthetic and exploratory. It does not estimate the real environmental impact of AI systems, data centers, or model inference. Instead, it investigates whether ecological constraints can be operationalized as endogenous governance signals inside an agentic AI architecture. The accompanying Jupyter notebook contains the executable experiment, exported figures, tables, and intervention logs. This report provides the conceptual synthesis and interpretation of that computational artifact.

1. Introduction

Contemporary AI systems are commonly optimized around capability, scale, throughput, latency, and performance. The underlying assumption is often implicit: additional computation is permissible unless explicit technical limits are reached. In this framing, recursion, inference, and escalation are treated as computational abstractions rather than materially situated processes.

Yet computation is inseparable from infrastructure. AI systems depend on energy production, thermal regulation, water usage, semiconductor supply chains, data center operations, and hardware manufacturing. Even relatively small inference workloads exist within broader planetary systems shaped by resource extraction, energy availability, environmental constraint, and thermodynamic cost.

PEMAS explores a different architectural assumption:

Planetary operating conditions can function as first-class governance signals inside an AI system.

The project therefore reframes ecological context as an operational component of cognition rather than an external reporting layer. It asks whether a multi-agent system can behave differently when ecological state is not merely measured after the fact, but placed inside the decision loop that regulates computational behavior.

This report should be read alongside the accompanying notebook. The notebook is the executable research artifact. It contains the synthetic simulation engine, the baseline and governed agent pipelines, the experiment code, the output tables, and the plotted figures. The report synthesizes the architecture, results, limitations, and conceptual contribution in a compact preprint-style format.

2. Research Question

The primary research question is:

Can a multi-agent AI system maintain useful task performance while dynamically adapting cognition to simulated planetary constraints?

This question can be decomposed into several subordinate questions. Can ecological governance reduce unnecessary recursive refinement? Does adaptive sufficiency alter computational waste? How does ecological constraint change model-routing behavior? Can planetary-aware governance reduce runaway or excessive multi-agent loops? More broadly, what forms of computational behavior emerge when cognition is treated as bounded by ecological operating context?

The project does not attempt to prove that PEMAS is universally superior to a baseline system. Instead, it investigates whether ecological constraints can be represented as operational governance signals and whether those signals change computational behavior in measurable and interpretable ways.

3. Conceptual Framing

PEMAS does not attempt to build a production-ready sustainability platform. Instead, it functions as an exploratory systems architecture experiment situated at the intersection of AI governance, multi-agent systems, bounded rationality, sustainable computing, cybernetics, and systems ecology.

The framework draws particularly from traditions of bounded rationality and cybernetic feedback, while extending these ideas toward ecologically situated computational governance. Bounded rationality is relevant because PEMAS does not assume that more reasoning is always better. Cybernetics is relevant because the architecture uses feedback from environmental state, task quality, uncertainty, and resource expenditure to alter system behavior.

The central conceptual shift introduced by PEMAS is that cognition becomes environmentally situated. Under this framing, computation is no longer treated as abstractly detached from material conditions. Instead, recursive reasoning, escalation, compression, routing behavior, and stopping decisions are regulated according to a simulated ecological operating context.

In this sense, PEMAS treats ecological state not as an external reporting layer, but as an endogenous component of the cognition loop itself. The architecture does not merely ask how much computation cost after the fact. It asks whether computation should continue, compress, reroute, or stop under the current operating conditions.

4. Experimental Framework

The experiment compares two architectural assumptions about computation. The baseline system represents a conventional multi-agent workflow in which recursive refinement proceeds according to task-quality logic alone. Computation is implicitly treated as available unless a quality threshold, critic decision, or maximum iteration count terminates the loop.

The PEMAS system introduces a different assumption: ecological operating conditions can function as endogenous governance signals inside the cognition loop. Both systems use the same synthetic task set, agent structure, and evaluation logic. The key experimental difference is whether ecological state is permitted to alter computational behavior.

The purpose of the experiment is therefore not to determine whether ecological governance is universally superior. Instead, the objective is to investigate how cognition changes when computational activity becomes environmentally situated.

4.1 Baseline System

The baseline pipeline represents a conventional simulated agentic workflow: a Planner Agent decomposes the task, a Worker Agent generates a response, and a Critic Agent evaluates quality and uncertainty. Recursive refinement continues while the quality threshold has not been reached, the critic recommends additional refinement, and the maximum iteration count has not been exceeded.

The baseline has no awareness of carbon intensity, water stress, thermal load, energy availability, hardware scarcity, or ecological load. This makes it a useful control condition. It captures the behavior of a system whose stopping conditions are defined only by task performance and internal refinement logic.

4.2 PEMAS System

The PEMAS pipeline introduces ecological governance into the cognition loop. It uses the same Planner, Worker, and Critic roles as the baseline, but adds a synthetic planetary-state engine, a Planetary Governor Agent, and a Sufficiency Agent.

The Planetary Governor evaluates the current ecological load and issues governance decisions. These decisions can reduce token budgets, restrict recursion, force compression, deny escalation, or route computation toward a smaller model configuration. The Sufficiency Agent evaluates whether further cognition is justified given current quality, uncertainty, marginal improvement, and ecological stress.

The architecture therefore modifies computational behavior through a structured governance loop rather than through a simple fixed cap. Under low stress, richer reasoning may be permitted. Under high stress, recursion may be restricted and compression may be required. Under moderate stress, the system may continue but with reduced budgets and stronger sufficiency pressure.

4.3 Shared Task Set and Metrics

Both systems operate on the same synthetic task set, including summarization, classification, structured reasoning, planning, scientific reasoning, and synthesis tasks. Each task includes a difficulty score and an ambiguity score. These values influence simulated token use, quality progression, and uncertainty.

The experiment evaluates final quality proxy, final uncertainty proxy, cumulative synthetic token use, iteration depth, stopping reason, mean ecological load, ecological cost proxy, routing behavior, and compression rate. It also introduces an exploratory Cognitive Ecological Efficiency metric, defined as final quality divided by the product of cumulative token use and iteration count. This metric is not proposed as universal. It is used only for internal comparison within the synthetic experiment.

5. Planetary-State Engine

The planetary-state engine generates synthetic ecological operating conditions. It includes carbon intensity, water stress, thermal load, energy availability, hardware scarcity, and a composite ecological load index. All variables are normalized between 0.0 and 1.0 to make the experiment interpretable and reproducible.

The ecological load index rises when carbon intensity, water stress, thermal load, and hardware scarcity increase, and when energy availability falls. This composite value functions as the primary ecological governance signal.

The notebook implements six synthetic ecological scenarios: low stress, moderate stress, high stress, drought, energy scarcity, and hardware scarcity. These scenarios are not empirical environmental measurements. They are controlled experimental regimes designed to test whether the same agentic system behaves differently under different operating conditions.

The notebook output shows that the synthetic trajectory behaves as intended: ecological pressure rises over time, energy availability falls under high-stress conditions, and the ecological load index tracks aggregate pressure. Scenario profiles also separate clearly, allowing later analysis of how governance behavior changes across stress regimes.

6. Experimental Findings

The notebook results show that ecological operating context changes computational behavior in measurable ways. The baseline continues refinement according to internal quality and uncertainty dynamics. PEMAS, by contrast, contracts or expands computation depending on the synthetic planetary state.

6.1 Token Expenditure

Token expenditure is one of the clearest differences between the baseline and PEMAS. In low-stress conditions, PEMAS permits richer cognition and therefore higher cumulative token use. Under high-stress, drought, energy-scarcity, and hardware-scarcity conditions, token expenditure contracts sharply. This demonstrates that ecological state is not merely being recorded. It actively governs computational behavior.

6.2 Recursive Depth

Recursive depth also changes across scenarios. Low-stress conditions allow longer refinement sequences, while severe stress conditions restrict additional iterations. This is an important result because recursive loops are one of the places where multi-agent systems can accumulate unnecessary computational expenditure. PEMAS operationalizes the idea that recursion should not be treated as automatically neutral or free.

6.3 Quality and Resource Trade-off

The quality-resource trade-off is the central empirical pattern of the experiment. PEMAS does not preserve quality uniformly across all ecological conditions. Instead, it allows higher quality and richer reasoning under low stress, while accepting degraded quality under severe constraint. This is not a failure of the framework. It is the trade-off under investigation.

The result shows that ecological governance can shift the system from unconditional capability maximization toward adaptive sufficiency. In moderate conditions, the system can still preserve useful performance while reducing computational expenditure. In extreme conditions, capability contracts more aggressively. This creates a visible spectrum between capability, restraint, and utility degradation.

6.4 Routing, Compression, and Stopping Behavior

The intervention logs show that PEMAS changes not only how much computation occurs, but why computation stops and how it is routed. Low-stress scenarios permit larger routing and deeper reasoning.

Higher-stress scenarios shift routing toward smaller configurations, force compression more often, and stop through governance or sufficiency conditions rather than ordinary target-quality completion.

This matters because it demonstrates that the architecture is not simply a token limiter. It is a governance system with multiple intervention types: routing, throttling, compression, recursion denial, and sufficiency-based termination. The resulting behavior is differentiated across ecological regimes.

6.5 Cognitive Ecological Efficiency

The exploratory Cognitive Ecological Efficiency metric provides an additional perspective on the trade-off between quality and expenditure. In severe stress conditions, CEE may increase because token use and recursive depth collapse faster than quality falls. This produces a methodological caution: a system can appear computationally efficient while becoming less useful.

This observation points toward an important future research question. Ecological governance cannot be reduced to minimizing computation. A well-designed system must balance computational restraint with retained utility, uncertainty management, and acceptable degradation. Otherwise, efficiency can become indistinguishable from premature shutdown.

7. Discussion

The key contribution of PEMAS is architectural rather than environmental. The framework demonstrates that ecological context can be represented as an endogenous governance signal inside a multi-agent AI system. This is different from merely reporting carbon use, energy use, or token counts after computation has occurred.

PEMAS reframes computation as environmentally situated. Under this framing, cognition does not float above planetary systems as an abstract optimization process. It operates inside a simulated ecological context that changes what forms of reasoning are permitted, how long refinement continues, and when sufficiency is declared.

The metaphor of bounded computational metabolism is useful here. A living organism does not maximize every process indefinitely. It allocates energy, conserves resources, shifts behavior under stress, and stops when marginal returns no longer justify expenditure. PEMAS applies a similar principle to agentic computation, not as biological analogy for its own sake, but as a systems design hypothesis.

The project also clarifies a broader governance issue. In many AI systems, the default assumption is that more reasoning, more recursion, and more escalation are preferable if they improve output quality. PEMAS introduces a counter-assumption: further cognition may need to be justified under environmental constraint. This does not make performance irrelevant. It changes the conditions under which performance improvement is pursued.

8. Limitations

The framework is intentionally synthetic and exploratory. The planetary variables are not real environmental measurements. The ecological load index is a proxy, not a scientific environmental impact model. The agents are simulated rather than LLM-backed. Token use, quality, uncertainty, and ecological cost are also simulated. The task set is small and illustrative rather than a benchmark suite.

The governance layer uses threshold-based heuristics. These thresholds are transparent and interpretable, but they are not learned, optimized, or validated against real infrastructure data. Future versions could explore learned governance policies, uncertainty-aware controllers, or adaptive ecological routing mechanisms.

The experiment also does not include human evaluation. The quality proxy is useful for demonstrating comparative behavior, but it should not be interpreted as a substitute for human assessment or domain-specific task evaluation.

These limitations do not invalidate the project. They define its scope. PEMAS v0.1 is an architectural feasibility experiment. Its purpose is to show that ecological state can be placed inside the cognition loop and can alter agentic behavior in reproducible ways.

9. Future Work

Future work could extend PEMAS in several directions. The most immediate extension would be to replace simulated workers with real LLM calls and real token accounting. This would allow the framework to evaluate whether similar governance patterns emerge under real inference workloads.

A second extension would incorporate real environmental data sources, such as carbon-intensity APIs, cloud-region information, electricity mix, water stress indicators, or thermal constraints. This would move the framework from synthetic ecological scenarios toward infrastructure-aware orchestration.

A third extension would involve ablation studies. Researchers could compare full PEMAS against versions without the Sufficiency Agent, without compression, without routing constraints, or without recursion control. This would clarify which governance components contribute most to computational restraint and retained utility.

Finally, future work should explore utility thresholds more carefully. The CEE metric revealed that computational efficiency can increase even when usefulness declines. A more mature framework would therefore require metrics that jointly consider ecological cost, quality retention, uncertainty, and minimum utility thresholds.

10. Conclusion

This report introduced PEMAS, a synthetic experimental framework for planetary-embedded multi-agent governance. The central contribution is architectural: planetary operating conditions can function as first-class governance signals inside an agentic AI system.

The accompanying notebook demonstrates that simulated ecological state can alter token expenditure, recursion depth, model routing, compression behavior, and stopping decisions. These results support the feasibility of treating cognition as bounded by an ecological operating context rather than as a computational process implicitly detached from material conditions.

The framework is intentionally synthetic and exploratory. It is not designed to estimate real environmental impact or provide deployment-ready governance policies. Its value lies in operationalizing ecological embeddedness inside a reproducible computational research artifact.

More broadly, PEMAS suggests a possible shift in how intelligent systems are conceptualized. Rather than assuming that cognition exists independently of planetary substrate, future architectures may increasingly need to operate as participants within constrained ecological systems. In this framing, computation is no longer treated as abstractly free, recursion is no longer environmentally neutral, and cognition itself becomes environmentally situated.

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Reproducibility Statement

All experiments described in this report are implemented in the accompanying executable Jupyter notebook. The notebook uses fixed random seeds and exports figures, tables, and intervention logs to an outputs directory. The report should therefore be read as the interpretive synthesis of a reproducible computational artifact rather than as a standalone empirical study.